

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 14B

ACID-BASE TITRATIONS, PH CURVES, INDICATORS & BUFFER SOLUTIONS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 14: ACID-BASE EQUILIBRIA
SUB-TOPIC:	14B: Acid-Base Titrations, pH Curves, Indicators & Buffer Solutions
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	203 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Titration pH Curve & Indicator Selection****[5 Marks]****WCH14/01/Jan23/Q13**

25.0 cm³ of 0.100 mol dm⁻³ CH₃COOH (pK_a = 4.76) is titrated with 0.100 mol dm⁻³ NaOH.

- (a) Sketch the pH curve for this titration from 0 to 40 cm³ of NaOH added, labelling key features. [3]
- (b) Select a suitable indicator from Phenolphthalein (pK_{in} = 9.3) and Methyl Orange (pK_{in} = 3.7). Justify your choice. [2]

Q2. 2. Buffer Action Mechanism**[4 Marks]****WCH14/01/Oct22/Q14**

An acidic buffer solution contains ethanoic acid (CH₃COOH) and sodium ethanoate (CH₃COONa).

- (a) Define a buffer solution. [2]
- (b) Explain how this buffer resists changes in pH when a small amount of acid (H⁺) is added. [2]

Q3. 3. Buffer pH Calculation Step 3**[3 Marks]****WCH14/01/Sample/Q3**

Buffer solution 3 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa (K_a = 1.74 × 10⁻⁵ mol dm⁻³).

- (a) Calculate [H⁺] in the buffer using K_a = [H⁺][CH₃COO⁻] / [CH₃COOH]. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q4. 4. Buffer pH Calculation Step 4**[3 Marks]****WCH14/01/Sample/Q4**

Buffer solution 4 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa (K_a = 1.74 × 10⁻⁵ mol dm⁻³).

- (a) Calculate [H⁺] in the buffer using K_a = [H⁺][CH₃COO⁻] / [CH₃COOH]. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q5. 5. Buffer pH Calculation Step 5**[3 Marks]****WCH14/01/Sample/Q5**

Buffer solution 5 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa (K_a = 1.74 × 10⁻⁵ mol dm⁻³).

- (a) Calculate [H⁺] in the buffer using K_a = [H⁺][CH₃COO⁻] / [CH₃COOH]. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q6. 6. Buffer pH Calculation Step 6**[3 Marks]****WCH14/01/Sample/Q6**

Buffer solution 6 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa (K_a = 1.74 × 10⁻⁵ mol dm⁻³).

- (a) Calculate [H⁺] in the buffer using K_a = [H⁺][CH₃COO⁻] / [CH₃COOH]. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q7. 7. Buffer pH Calculation Step 7**[3 Marks]****WCH14/01/Sample/Q7**

Buffer solution 7 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa (K_a = 1.74 × 10⁻⁵ mol dm⁻³).

- (a) Calculate [H⁺] in the buffer using K_a = [H⁺][CH₃COO⁻] / [CH₃COOH]. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q8. 8. Buffer pH Calculation Step 8**[3 Marks]****WCH14/01/sample/Q8**

Buffer solution 8 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q9. 9. Buffer pH Calculation Step 9**[3 Marks]****WCH14/01/sample/Q9**

Buffer solution 9 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q10. 10. Buffer pH Calculation Step 10**[3 Marks]****WCH14/01/sample/Q10**

Buffer solution 10 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q11. 11. Buffer pH Calculation Step 11**[3 Marks]****WCH14/01/sample/Q11**

Buffer solution 11 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q12. 12. Buffer pH Calculation Step 12**[3 Marks]****WCH14/01/sample/Q12**

Buffer solution 12 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q13. 13. Buffer pH Calculation Step 13**[3 Marks]****WCH14/01/sample/Q13**

Buffer solution 13 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q14. 14. Buffer pH Calculation Step 14**[3 Marks]****WCH14/01/sample/Q14**

Buffer solution 14 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

[3 Marks]**Q15. 15. Buffer pH Calculation Step 15****WCH14/01/sample/Q15**

Buffer solution 15 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

[3 Marks]**Q16. 16. Buffer pH Calculation Step 16****WCH14/01/sample/Q16**

Buffer solution 16 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

[3 Marks]**Q17. 17. Buffer pH Calculation Step 17****WCH14/01/sample/Q17**

Buffer solution 17 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

[3 Marks]**Q18. 18. Buffer pH Calculation Step 18****WCH14/01/sample/Q18**

Buffer solution 18 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

[3 Marks]**Q19. 19. Buffer pH Calculation Step 19****WCH14/01/sample/Q19**

Buffer solution 19 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

[3 Marks]**Q20. 20. Buffer pH Calculation Step 20****WCH14/01/sample/Q20**

Buffer solution 20 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

[3 Marks]**Q21. 21. Buffer pH Calculation Step 21****WCH14/01/sample/Q21**

Buffer solution 21 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q22. 22. Buffer pH Calculation Step 22**[3 Marks]****WCH14/01/Sample/Q22**

Buffer solution 22 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q23. 23. Buffer pH Calculation Step 23**[3 Marks]****WCH14/01/Sample/Q23**

Buffer solution 23 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q24. 24. Buffer pH Calculation Step 24**[3 Marks]****WCH14/01/Sample/Q24**

Buffer solution 24 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

Q25. 25. Buffer pH Calculation Step 25**[3 Marks]****WCH14/01/Sample/Q25**

Buffer solution 25 contains 0.200 mol dm⁻³ CH₃COOH and 0.300 mol dm⁻³ CH₃COONa ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³).

- (a) Calculate $[H^+]$ in the buffer using $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$. [2]
- (b) Calculate the pH of the buffer solution. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 26****[5 Marks]**
WCH14/01/Hard/Q26

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q27. 27. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 27**[5 Marks]**
WCH14/01/Hard/Q27

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q28. 28. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 28**[5 Marks]**
WCH14/01/Hard/Q28

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q29. 29. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 29**[5 Marks]**
WCH14/01/Hard/Q29

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q30. 30. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 30**[5 Marks]**
WCH14/01/Hard/Q30

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q31. 31. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 31**[5 Marks]**
WCH14/01/Hard/Q31

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q32. 32. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 32**[5 Marks]**
WCH14/01/Hard/Q32

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH_3COOH and CH_3COO^- after reaction with added NaOH . [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q33. 33. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 33**

[5 Marks]
WCH14/01/Hard/Q33

A 500 cm³ buffer solution contains 0.100 mol CH_3COOH and 0.100 mol CH_3COONa ($K_a = 1.74 \times 10^{-5}$, $\text{p}K_a = 4.76$).
0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH_3COOH and CH_3COO^- after reaction with added NaOH . [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q34. 34. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 34**

[5 Marks]
WCH14/01/Hard/Q34

A 500 cm³ buffer solution contains 0.100 mol CH_3COOH and 0.100 mol CH_3COONa ($K_a = 1.74 \times 10^{-5}$, $\text{p}K_a = 4.76$).
0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH_3COOH and CH_3COO^- after reaction with added NaOH . [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q35. 35. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 35**

[5 Marks]
WCH14/01/Hard/Q35

A 500 cm³ buffer solution contains 0.100 mol CH_3COOH and 0.100 mol CH_3COONa ($K_a = 1.74 \times 10^{-5}$, $\text{p}K_a = 4.76$).
0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH_3COOH and CH_3COO^- after reaction with added NaOH . [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q36. 36. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 36**

[5 Marks]
WCH14/01/Hard/Q36

A 500 cm³ buffer solution contains 0.100 mol CH_3COOH and 0.100 mol CH_3COONa ($K_a = 1.74 \times 10^{-5}$, $\text{p}K_a = 4.76$).
0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH_3COOH and CH_3COO^- after reaction with added NaOH . [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q37. 37. A* Challenge: Buffer pH After Addition of Acid/Alkali &
Half-Equivalence 37**

[5 Marks]
WCH14/01/Hard/Q37

A 500 cm³ buffer solution contains 0.100 mol CH_3COOH and 0.100 mol CH_3COONa ($K_a = 1.74 \times 10^{-5}$, $\text{p}K_a = 4.76$).
0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH_3COOH and CH_3COO^- after reaction with added NaOH . [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q38. 38. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 38**

[5 Marks]
WCH14/01/Hard/Q38

A 500 cm³ buffer solution contains 0.100 mol CH_3COOH and 0.100 mol CH_3COONa ($K_a = 1.74 \times 10^{-5}$, $\text{p}K_a = 4.76$).
0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH_3COOH and CH_3COO^- after reaction with added NaOH . [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q39. 39. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 39**

[5 Marks]
WCH14/01/Hard/Q39

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q40. 40. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 40

[5 Marks]
WCH14/01/Hard/Q40

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q41. 41. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 41

[5 Marks]
WCH14/01/Hard/Q41

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q42. 42. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 42

[5 Marks]
WCH14/01/Hard/Q42

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q43. 43. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 43

[5 Marks]
WCH14/01/Hard/Q43

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q44. 44. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 44

[5 Marks]
WCH14/01/Hard/Q44

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q45. 45. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 45

[5 Marks]
WCH14/01/Hard/Q45

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

Q46. 46. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 46

[5 Marks]
WCH14/01/Hard/Q46

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q47. 47. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 47**

[5 Marks]
WCH14/01/Hard/Q47

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q48. 48. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 48**

[5 Marks]
WCH14/01/Hard/Q48

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q49. 49. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 49**

[5 Marks]
WCH14/01/Hard/Q49

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

**Q50. 50. A* Challenge: Buffer pH After Addition of Acid/Alkali
& Half-Equivalence 50**

[5 Marks]
WCH14/01/Hard/Q50

A 500 cm³ buffer solution contains 0.100 mol CH₃COOH and 0.100 mol CH₃COONa ($K_a = 1.74 \times 10^{-5}$, $pK_a = 4.76$). 0.010 mol of solid NaOH is added without changing the volume.

- (a) Calculate the new moles of CH₃COOH and CH₃COO⁻ after reaction with added NaOH. [2]
(b) Calculate the new pH of the buffer solution. [3]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 14B

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Titration pH Curve & Indicator Selection 1. (a) Initial pH ~2.9, buffer region around pH 4.8, steep vertical section at 25.0 cm ³ (pH 7–10), final pH ~12.5 (3). 1. (b) Phenolphthalein (1). Its pK _{in} (9.3) / color change range (8.3–10.0) lies entirely within the vertical section of the titration curve (pH 7–10) (1).	[5]
Q2	Q2: 2. Buffer Action Mechanism 2. (a) A solution that resists changes in pH when small amounts of acid or base/alkali are added (2). 2. (b) Added H ⁺ ions react with large reservoir of ethanoate ions: CH ₃ COO ⁻ + H ⁺ → CH ₃ COOH (1). Keeps [H ⁺] nearly constant (1).	[4]
Q3	Q3: 3. Buffer pH Calculation Step 3 3. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 3. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q4	Q4: 4. Buffer pH Calculation Step 4 4. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 4. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q5	Q5: 5. Buffer pH Calculation Step 5 5. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 5. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q6	Q6: 6. Buffer pH Calculation Step 6 6. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 6. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q7	Q7: 7. Buffer pH Calculation Step 7 7. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 7. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q8	Q8: 8. Buffer pH Calculation Step 8 8. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 8. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q9	Q9: 9. Buffer pH Calculation Step 9 9. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 9. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q10	Q10: 10. Buffer pH Calculation Step 10 10. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 10. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q11	Q11: 11. Buffer pH Calculation Step 11 11. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 11. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q12	Q12: 12. Buffer pH Calculation Step 12 12. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 12. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]

Q13	Q13: 13. Buffer pH Calculation Step 13 13. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 13. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q14	Q14: 14. Buffer pH Calculation Step 14 14. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 14. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q15	Q15: 15. Buffer pH Calculation Step 15 15. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 15. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q16	Q16: 16. Buffer pH Calculation Step 16 16. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 16. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q17	Q17: 17. Buffer pH Calculation Step 17 17. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 17. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q18	Q18: 18. Buffer pH Calculation Step 18 18. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 18. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q19	Q19: 19. Buffer pH Calculation Step 19 19. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 19. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q20	Q20: 20. Buffer pH Calculation Step 20 20. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 20. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q21	Q21: 21. Buffer pH Calculation Step 21 21. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 21. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q22	Q22: 22. Buffer pH Calculation Step 22 22. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 22. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q23	Q23: 23. Buffer pH Calculation Step 23 23. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 23. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q24	Q24: 24. Buffer pH Calculation Step 24 24. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 24. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]
Q25	Q25: 25. Buffer pH Calculation Step 25 25. (a) $[H^+] = K_a \times [CH_3COOH] / [CH_3COO^-] = (1.74 \times 10^{-5})(0.200) / (0.300) = 1.16 \times 10^{-5} \text{ mol dm}^{-3}$ (2). 25. (b) $pH = -\log(1.16 \times 10^{-5}) = 4.94$ (1).	[3]

Q26	Q26: 26. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 26 26. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 26. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q27	Q27: 27. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 27 27. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 27. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q28	Q28: 28. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 28 28. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 28. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q29	Q29: 29. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 29 29. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 29. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q30	Q30: 30. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 30 30. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 30. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q31	Q31: 31. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 31 31. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 31. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q32	Q32: 32. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 32 32. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 32. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q33	Q33: 33. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 33 33. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 33. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q34	Q34: 34. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 34 34. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 34. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]

Q35	Q35: 35. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 35 35. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 35. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q36	Q36: 36. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 36 36. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 36. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q37	Q37: 37. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 37 37. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 37. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q38	Q38: 38. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 38 38. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 38. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q39	Q39: 39. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 39 39. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 39. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q40	Q40: 40. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 40 40. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 40. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q41	Q41: 41. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 41 41. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 41. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q42	Q42: 42. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 42 42. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 42. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q43	Q43: 43. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 43 43. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 43. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]

Q44	Q44: 44. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 44 44. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 44. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q45	Q45: 45. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 45 45. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 45. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q46	Q46: 46. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 46 46. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 46. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q47	Q47: 47. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 47 47. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 47. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q48	Q48: 48. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 48 48. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 48. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q49	Q49: 49. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 49 49. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 49. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]
Q50	Q50: 50. A* Challenge: Buffer pH After Addition of Acid/Alkali & Half-Equivalence 50 50. (a) Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. New moles CH ₃ COOH = 0.100 - 0.010 = 0.090 mol (1). New moles CH ₃ COO ⁻ = 0.100 + 0.010 = 0.110 mol (1). 50. (b) [H ⁺] = K _a x (moles HA / moles A ⁻) = (1.74x10 ⁻⁵)(0.090 / 0.110) = 1.424 x 10 ⁻⁵ mol dm ⁻³ (2). New pH = -log(1.424x10 ⁻⁵) = 4.85 (1).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 14B

FAQ #1: Half-Equivalence Point Property

Category: Titration Curve
Feature

COMMON EXAMINER TRAP / MISTAKE:

Forgetting that $\text{pH} = \text{pK}_a$ at the half-equivalence point.

MODEL ANSWER / EXAMINER EXPECTATION:

At the half-equivalence point in a weak acid-strong base titration, exactly half of the weak acid has been neutralised ($[\text{HA}] = [\text{A}^-]$). Therefore $\text{K}_a = [\text{H}^+]$ and $\text{pH} = \text{pK}_a$.

FAQ #2: Indicator Selection Criterion

Category: Indicator Range

COMMON EXAMINER TRAP / MISTAKE:

Selecting an indicator based solely on colour preference.

MODEL ANSWER / EXAMINER EXPECTATION:

An indicator is suitable ONLY if its pH range ($\text{pK}_{\text{In}} \pm 1$) falls ENTIRELY within the steep vertical section of the titration pH curve.

FAQ #3: Weak Acid-Weak Base Indicator Suitability

Category: Titration Curves

COMMON EXAMINER TRAP / MISTAKE:

Selecting indicators for weak acid-weak base titrations.

MODEL ANSWER / EXAMINER EXPECTATION:

Weak acid – weak base titrations have NO steep vertical section on their pH curve. Therefore NO indicator is suitable; a pH meter must be used.

FAQ #4: Buffer Calculation Moles vs Concentration

Category: Buffer Shortcuts

COMMON EXAMINER TRAP / MISTAKE:

Thinking volume must always be used when calculating buffer pH.

MODEL ANSWER / EXAMINER EXPECTATION:

In $\text{K}_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$, volume cancels out because $[\text{A}^-]$ and $[\text{HA}]$ are in the same total volume. You can calculate $[\text{H}^+] = \text{K}_a \times \frac{\text{moles HA}}{\text{moles A}^-}$ directly.

FAQ #5: Addition of Acid to Buffer Mechanism

Category: Buffer Action

COMMON EXAMINER TRAP / MISTAKE:

Writing H^+ reacts with HA instead of A^- .

MODEL ANSWER / EXAMINER EXPECTATION:

Added H^+ reacts with the CONJUGATE BASE A^- : $\text{A}^- + \text{H}^+ \rightarrow \text{HA}$. Added OH^- reacts with the WEAK ACID HA: $\text{HA} + \text{OH}^- \rightarrow \text{A}^- + \text{H}_2\text{O}$.

FAQ #6: Basic Buffer Equations

Category: Basic Buffers

COMMON EXAMINER TRAP / MISTAKE:

Using K_a equations directly for basic buffers like $\text{NH}_3/\text{NH}_4^+$.

MODEL ANSWER / EXAMINER EXPECTATION:

For basic buffers ($\text{NH}_3/\text{NH}_4^+$): use K_b or K_a of NH_4^+ ($K_a = K_w / K_b$). Alternatively: $[\text{OH}^-] = K_b \times [\text{NH}_3]/[\text{NH}_4^+]$, then find $[\text{H}^+]$ via K_w .

FAQ #7: Blood Buffer System**Category: Physiological Buffers****COMMON EXAMINER TRAP / MISTAKE:**

Misidentifying the primary blood buffer components.

MODEL ANSWER / EXAMINER EXPECTATION:

The primary blood buffer is the carbonic acid – hydrogencarbonate system: $\text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$. CO_2 in lungs regulates $[\text{H}_2\text{CO}_3]$ while kidneys regulate $[\text{HCO}_3^-]$.

FAQ #8: Why Salt Is Needed in Acidic Buffers**Category: Buffer Formulation****COMMON EXAMINER TRAP / MISTAKE:**

Assuming a weak acid alone can act as a buffer.

MODEL ANSWER / EXAMINER EXPECTATION:

A weak acid dissociates poorly and provides very few A^- ions. Adding the sodium salt (NaA) provides a LARGE RESERVOIR of conjugate base A^- needed to neutralize added H^+ .

FAQ #9: Equivalence Point pH for Weak Acid–Strong Base**Category: Equivalence pH****COMMON EXAMINER TRAP / MISTAKE:**

Assuming the equivalence point is always pH 7.00.

MODEL ANSWER / EXAMINER EXPECTATION:

At the equivalence point of a weak acid – strong base titration, the solution contains the conjugate base A^- which hydrolyses water: $\text{A}^- + \text{H}_2\text{O} \rightleftharpoons \text{HA} + \text{OH}^-$. Thus $\text{pH} > 7$ (alkaline equivalence point).

FAQ #10: Equivalence Point pH for Strong Acid–Weak Base**Category: Equivalence pH****COMMON EXAMINER TRAP / MISTAKE:**

Assuming strong acid–weak base titration has alkaline equivalence point.

MODEL ANSWER / EXAMINER EXPECTATION:

At the equivalence point of a strong acid – weak base titration, the solution contains the conjugate acid BH^+ which hydrolyses water: $\text{BH}^+ + \text{H}_2\text{O} \rightleftharpoons \text{B} + \text{H}_3\text{O}^+$. Thus $\text{pH} < 7$ (acidic equivalence point).